

Pick and Protect Fragile Glassware SOP

One episode empties the pick row into one tote. The cell divider goes into the tote first. Then, for every glass, a bubble wrap sheet lines the next cell and the glass goes into that cell lying flat on its side, heaviest to lightest. After the last glass, one void pad fills the gap down each side of the divider, and the fragile sticker goes on the tote last. The episode ends when the pick row is empty and the tote is stickered.

The order never changes: divider, then sheet and glass, sheet and glass, again and again until the pick row is empty, then the pad-side void, then the other void, then the fragile sticker. No sheet goes in before the divider is seated on the tote floor. No glass goes into a cell before its sheet is lining it. No void pad goes in until the pick row is empty, and the fragile sticker does not go on until both voids are filled.

Each thing is taken by the gripper on its own side of the surface. The gripper on the divider's side sets the divider. The gripper on the sheets' side lines every cell. The gripper on the glasses' side takes every glass. The gripper on the pads' side fills the void on its own side, then hands the other pad to the other gripper, which fills the other void. The right gripper picks up the fragile sticker and both grippers stick it on together. No arm reaches across the surface.

Glass breaks when it is dropped or forced. A glass may be held and carried any way that works, as long as it is lowered into its cell and rests there lying flat on its side before the gripper opens. A glass never touches another glass, never touches bare cardboard, and is never dropped or forced.

In every step, either gripper or both may adjust anything that goes into the tote and push it down until it sits right: the divider, each sheet, each glass (gently), each void pad, and the sticker on the ledge. The tote and the pick row stay where they are put.

The surface is set up in one of eight ways. The tote stays at front center and the sticker spot at the front center, in front of the tote, in all eight. The divider, the void pads, the glasses, and the bubble wrap sheets start at the left-center, the right-center, the front left, and the front right, as the config sets. In Config 5 and Config 8 the pads lie beside the divider on the same spot.

- **Config 1:** the divider is at the **left-center** and the pads at the **front left**; the glasses are at the **right-center** and the sheets at the **front right**.
- **Config 2:** the divider is at the **left-center** and the pads at the **front right**; the glasses are at the **right-center** and the sheets at the **front left**.
- **Config 3:** the divider is at the **right-center** and the pads at the **front left**; the glasses are at the **left-center** and the sheets at the **front right**.
- **Config 4:** the divider is at the **right-center** and the pads at the **front right**; the glasses are at the **left-center** and the sheets at the **front left**.

- **Config 5:** the divider is at the **front left** and the pads at the **front left**; the glasses are at the **left-center** and the sheets at the **front right**.
- **Config 6:** the divider is at the **front left** and the pads at the **front right**; the glasses are at the **left-center** and the sheets at the **right-center**.
- **Config 7:** the divider is at the **front right** and the pads at the **front left**; the glasses are at the **left-center** and the sheets at the **right-center**.
- **Config 8:** the divider is at the **front right** and the pads at the **front right**; the glasses are at the **left-center** and the sheets at the **right-center**.

Where a step depends on the setup it says so on an **IF** line. Check the surface and follow the line that matches.

What stays constant across all sessions:

- **Start position:** one of the eight configs per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the divider's side sets it, the gripper on the sheets' side lines every cell, the gripper on the glasses' side takes every glass, and the gripper on the pads' side picks up both pads. No arm reaches across the surface.
- **Fixed roles:** the pad gripper hands the other pad over at the hand-over place, the right gripper picks up the fragile sticker, and both grippers stick it on together, in every config.
- **Order:** divider, then sheet and glass for every glass, then the pad-side void, then the other void, then the sticker, in every config.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole surface: the divider, the void pads, the pick row, and the sheet row where this episode's config puts them, the sticker spot at the front center, and the tote at front center.
3. The whole pick row is in frame, so it can be seen going empty, and every cell of the divider is visible from above once it is in the tote, so every glass can be seen going in.
4. Both arms are at home with grippers open.

5. The surface is bare apart from the glasses, the tote, the divider, the bubble wrap sheets, the two void pads, and the fragile sticker. There are no marks, mats, or fixtures on it.
6. Nothing obstructs the paths between each zone and the tote. Each arm reaches the center and the front of its own side of the surface, the hand-over place, and every cell of the tote. The right arm also reaches the sticker spot, and both arms reach the whole label ledge.

Materials checklist

1. Glasses stand upright in one row, the **pick row**, where this episode's config puts it, each on its own base with a clear gap to its neighbors, so a gripper closes on one without touching another. They are clean, dry, and free of chips and cracks.
2. The pick row holds glasses of visibly different weights, so **heavy glasses**, **medium glasses**, and **light glasses** are told apart at a glance by shape. Glasses of the same weight are identical to each other.
3. The pick row holds exactly as many glasses as the divider has cells, so no cell is left empty.
4. One open top **tote** at front center, square to the front edge, empty and dry. Its walls are low enough that a glass is lowered in from above without the gripper touching the rim. It has a flat **label ledge** along the top of its near wall.
5. One **cell divider**: a stiff cardboard grid, two cells wide and several rows deep, cut to drop into the tote snug against the front and back walls, with a gap down the left side wall and a gap down the right side wall. It stands on its own, open cells up, where this episode's config puts it. Each cell runs front to back, long enough for the tallest glass lying on its side and wide enough for the widest glass lying on its side with the open gripper down beside it. The walls are taller than the widest glass lying down.
6. **Bubble wrap sheets** stand in a row, the **sheet row**, where this episode's config puts them, one more sheet than there are glasses in the pick row. Each sheet is bubble wrap folded into an open square sleeve, open at the top and the bottom, the size of one cell and as tall as its walls. It stands on its own on a flat surface.
7. Two **void pads** lie side by side on the bare surface where this episode's config puts them, the **left void pad** on the left and the **right void pad** on the right, each lengthwise front to back. Each is a loosely crumpled roll of kraft paper as long as the tote's side wall, a little wider than the gap it fills, and soft enough to squash down as it goes in.
8. One printed **fragile sticker**, sticky side down on a strip of backing paper taped flat at the **sticker spot** at the front center, between the tote and the front edge of the surface. Its right end is folded back on itself as a dry **tab**.

Workspace layout

Every zone is a position on the bare surface. Nothing is marked, so each position is held by where the thing was set.

- **Divider spot:** where the config puts the divider: left-center (Config 1, 2), right-center (Config 3, 4), front left (Config 5, 6), or front right (Config 7, 8).
- **Pad spot:** where the config puts the two void pads: front left (Config 1, 3, 5, 7) or front right (Config 2, 4, 6, 8). In Config 5 and 8 the pads lie beside the divider on the same spot.
- **Pick row:** where the config puts the glasses: right-center (Config 1, 2) or left-center (Config 3 to 8).
- **Sheet row:** where the config puts the bubble wrap sheets: front right (Config 1, 3, 5), front left (Config 2, 4), or right-center (Config 6, 7, 8).
- **Sticker spot:** the front center, between the tote and the front edge of the surface.
- **Hand-over place:** in the air over the middle of the tote's near wall, just above the rim.
- **Tote:** front center, square to the front edge, in reach of both arms. It stays on the surface for the whole episode.

Each thing goes into the tote from the side it is picked on.

Vocabulary

- **Arm assignment:** each thing is taken by the gripper on its own side of the surface. The **divider gripper** is the gripper on the divider's side and sets the divider. The **sheet gripper** is the gripper on the sheets' side and lines every cell. The **glass gripper** is the gripper on the glasses' side and takes every glass. The **pad gripper** is the gripper on the pads' side; it picks up both void pads, fills the void on its own side, and hands the other pad to the **other gripper**, which fills the other void. The right gripper picks up the fragile sticker and both grippers stick it on together, in every config. One gripper often holds more than one role. Neither arm takes the other's zones.
- **Glass side:** the side the pick row starts on: right in Config 1 and 2, left in Config 3 to 8.
- **Pick row:** the glasses standing in a row at the start of the episode, at the left-center or right-center on the glass side.
- **Heavy glass:** a thick walled glass, short and wide on a heavy base. Heavy glasses go in first.
- **Medium glass:** a straight sided glass, thinner walled and taller than a heavy glass. Medium glasses go in after the heavy ones.
- **Light glass:** a thin walled glass, the tallest and narrowest kind. Light glasses go in last.
- **Weight order:** heaviest at the tote's back wall, lightest at the front. Each time, the heaviest glass still standing in the pick row is the next one in. When two glasses in the pick row weigh the same, take the one nearer the tote first.
- **Cell divider:** the cardboard grid that sits on the tote floor and gives every glass its own cell, so no glass can reach another.
- **Seated:** the divider rests flat on the tote floor, square to the tote walls, every cell open and square, no wall bent or folded over.

- **Next cell:** the empty cell in the backmost row that still has one. In that row, the cell away from the glass side goes first, then the cell on the glass side.
- **Bubble wrap sheet:** the folded bubble wrap sheet that lines a cell before its glass goes in, so the glass touches bubble wrap and never bare cardboard.
- **Lined:** the sheet rests on the tote floor inside its cell, open end up, flat against all four cell walls, with no side folded in, and stays that way with no gripper on it.
- **Void:** the gap between the divider and a tote side wall. There are two: the **left void** and the **right void**.
- **Pad-side void:** the void on the pad side. The **other void** is the one on the other side.
- **Void pad:** a crumpled paper roll that fills one void. The left void pad fills the left void, the right void pad fills the right void.
- **Filled:** the void pad sits down in the gap along its whole length, level with or below the top of the divider, with no tote floor visible in that gap, and no glass or divider moved by it going in.
- **Hand-over:** the pad gripper holds the other void pad still at the hand-over place, the other gripper closes on the middle of the pad, both hold for half a second, then the pad gripper opens and moves clear.
- **Lower in:** carrying the thing level over its spot, lowering it straight down until it is resting, and letting go once it is resting. Nothing is ever dropped in from above.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, tipping, or leaning.
- **Lying flat:** the glass lies on its side along the floor of its lined cell, not rolling and not propped up on its rim or its base, not leaning on the sheet or the cell wall.
- **Pick row empty:** no glass is left standing on the surface where the pick row started.
- **Release point:** over the spot in the tote, or over the label ledge, that the thing in the gripper is going to. A gripper opens nowhere else.
- **Label ledge:** the flat top of the tote's near wall, where the fragile sticker goes printed side up, square to the front edge and clear of the tote's opening.
- **Right way up:** the sticker lies printed side up with its words reading left to right from the front of the surface, not upside down and not turned on its side.

Steps

Steps 1 to 5 are one episode and one tote. A glass may be held and carried any way that works; what matters is that it is lowered in and rests lying flat on its side.

Steps 1, 2, and 3 depend on the config: each thing is taken by the gripper on its side, so the IF lines say which gripper that is. Step 4 and Step 5 are the same in all eight configs.

Step 1: Set the divider in the tote

Goal: the cell divider seated on the tote floor.

- **IF Config 1, 2, 5, or 6:** the **left gripper** pinches the top of the **divider's** middle wall. **IF Config 3, 4, 7, or 8:** the **right gripper** does.
- The same gripper lifts it straight up, carries it level to the **tote**, lowers it straight down until it rests on the tote floor, and lets go.
- Either gripper or both may adjust the divider and push it down into the tote until it is **seated**.

Check: the gripper is clear and the divider is **seated**. If it sits crooked or rides up on a tote wall, either gripper or both push it down or adjust it, or the same gripper lifts it clear and sets it again.

Step 2: Line a cell and nest a glass

Goal: the **pick row empty** and every glass **lying flat** in its own lined cell in **weight order**.

If the **pick row** is empty, go to Step 3. If a glass is still standing in it, run 2.1 and 2.2, then read this line again. Finish both substeps for one glass before starting the next glass.

2.1 Line the cell

- **IF Config 1, 3, 5, 6, 7, or 8:** the **right gripper** pinches the top edge of the next **bubble wrap sheet** in the sheet row. **IF Config 2 or 4:** the **left gripper** does.
- The same gripper lifts it straight up, carries it level to the **tote**, lowers it straight down into the **next cell** until it rests on the tote floor, and lets go once it is **lined**.
- Either gripper or both may push the sheet down and adjust it until it is **lined**.

Check: the gripper is clear and the cell is **lined**. If a side has folded in or the sheet is caught on a wall, either gripper or both push it down or adjust it, or the same gripper lifts it clear and lowers it in again.

2.2 Nest the glass

- Check the **pick row** and find the heaviest glass still standing in it. When two weigh the same, take the one nearer the tote.
- **IF Config 1 or 2:** the **right gripper** picks up that glass. **IF Config 3 to 8:** the **left gripper** does. It may hold and carry the glass any way that works.
- The same gripper **lowers** the glass **in** inside the sheet, lying flat on its side, until it rests on the tote floor, and lets go once it is resting.
- If the glass does not go all the way down, either gripper or both adjust it and push it gently down until it fits inside the sheet and rests lying flat on the tote floor.

Take the glass from whichever spot it stands in on the pick row. Never move another glass aside to reach it. Never pack a glass that is chipped, cracked, or broken, or one that cracks while it goes in.

Check: the gripper is clear and the glass is **lying flat** and **settled** in its lined cell, touching no bare cardboard. If it rolls, rocks, or sits propped up, either gripper or both adjust it and push it gently down, or the same gripper picks it up again and lowers it in again on its side.

Expected state after Step 2: the pick row is empty. Every glass lies flat on its side in its own lined cell, heaviest at the back wall and lightest at the front, and no glass touches another glass or bare cardboard.

Step 3: Pad the voids

Goal: both voids filled, with no glass or divider moved.

3.1 Fill the pad-side void

- **IF Config 1, 3, 5, or 7:** the **left gripper** pinches the **left void pad** at its middle. **IF Config 2, 4, 6, or 8:** the **right gripper** pinches the **right void pad** at its middle.
- The same gripper lifts it straight up, carries it level, lengthwise front to back, to above the void on its own side, lowers it straight down until it is **filled**, then lets go.
- Either gripper or both may adjust the pad and push it down into the gap until the void is **filled**.

Check: the gripper is clear and the pad-side void is **filled**. If the pad rides up or has moved the divider, either gripper or both push it down or adjust it, or the same gripper lifts it out and feeds it down again.

3.2 Pass and Fill the other void pad

- **IF Config 1, 3, 5, or 7:** the **left gripper** pinches the **right void pad** near its back end. **IF Config 2, 4, 6, or 8:** the **right gripper** pinches the **left void pad** near its back end.
- The same gripper lifts it straight up, carries it level, lengthwise front to back, to the **hand-over place**, and holds it still while the other gripper comes in and closes on the middle of the pad.
- Both grippers hold for half a second, then the pad gripper lets go and moves clear to its own side.
- The pad now hangs from the other gripper alone, still lengthwise front to back, and the pad gripper is clear. If the pad drops, the pad gripper picks it up where it landed and passes it again.
- **IF Config 1, 3, 5, or 7:** the **right gripper** carries the pad level to above the **right void**. **IF Config 2, 4, 6, or 8:** the **left gripper** carries it level to above the **left void**.
- The same gripper lowers it straight down until the void is **filled**, then lets go.
- Either gripper or both may adjust the pad and push it down into the gap until the void is **filled**.

Check: both grippers are clear of the tote and the tote is in frame. No tote floor shows down either side of the divider, every glass still lies **flat**, and neither pad rides above the top of the divider. If a pad rides up, either gripper or both push it down or adjust it. If a pad has moved a glass, either gripper or both adjust the glass until it lies flat again.

Step 4: Stick the fragile sticker

Goal: the fragile sticker stuck **right way up** on the tote's label ledge.

- The **right gripper** pinches the **tab** of the **fragile sticker** at the front center and lifts it straight up until the sticker peels free of the backing.
- The **right gripper** carries it level, printed side up, to just above the **label ledge**, without turning it.
- The **left gripper** pinches the sticker's left end. Both grippers hold it flat, **right way up** and square to the front edge.
- Both grippers lower it together flat onto the label ledge and let go once it is resting.
- Both grippers close and press it down together, the **left gripper** on the left half and the **right gripper** on the right half, so it sticks flat.
- Either gripper or both may adjust the sticker and press it again until it sits flat and square.

The sticker goes on only after both voids are filled. Never stick it inside the tote, on a glass, or on a void pad.

Check: both grippers are clear and the sticker lies flat, stuck, and **right way up** on the label ledge, readable in frame, square to the front edge, and clear of the tote's opening. If it sits crooked, either gripper or both adjust it and press it again. If it hangs off the ledge, lands folded, or is not the right way up, the **right gripper** peels it up by its tab and both grippers lay it again.

Step 5: End the episode

Check the tote once: the divider seated, every glass lying flat in its own lined cell in weight order, both voids filled, and the sticker on the label ledge, with the pick row empty. Return both arms home, clear of the tote and the pick row, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Peel the fragile sticker off the label ledge and lay it back sticky side down on its backing at the sticker spot, with the tab folded back. Replace it when it no longer sticks.
2. Lift the two void pads out of the tote and lay them side by side at the next episode's pad spot, lengthwise front to back, the left pad on the left. Recrumple any pad that has gone flat, and replace one that is limp or torn.

3. Lift the glasses out one at a time, front to back, and set them upright in a row at the next episode's pick row spot, with a clear gap between neighbors.
4. Lift the bubble wrap sheets out of the cells and stand them back in a row at the next episode's sheet row spot, with the sheets that were not used. Refold any sheet that no longer stands on its own, and replace one that is torn or has lost its bubbles. Confirm the sheet row holds one more sheet than there are glasses now in the pick row.
5. Lift the divider out and stand it at the next episode's divider spot. Replace it if a wall is bent, torn, or no longer square.
6. Set the empty tote back at front center, square to the front edge, and wipe out any paper scraps.
7. Vary which glass stands where in the pick row between episodes rather than rebuilding the same row every time, and vary the mix of weights in it. The pick order stays heaviest first and lightest last.
8. Confirm the pick row still holds exactly as many glasses as the divider has cells.
9. Check every glass in the light: replace any glass that is chipped, cracked, starred, or will no longer stand on its own base. A chipped glass never goes back in the pick row.
10. Wipe the surface and confirm it is dry and bare. A slick, wet, or crumbed surface lets a glass and the tote slide.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the surface: the divider, the pads, the glasses, or the sheets are not where the config puts them; a gripper reaches across the surface for something on the other side; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (each thing is taken by the gripper on its side; no arm reaches across the surface; the IF line followed is the one for the

config on the surface).

- **Coaching note:** check where the divider, the pads, the glasses, and the sheets are before the first reach, then follow that config's IF lines through Steps 1, 2, and 3.

Violation: Wrong order

- **Visible cue:** a sheet goes in before the divider is seated, a glass goes into a cell before its sheet is lining it, a void pad goes in while a glass is still standing in the pick row, or the fragile sticker goes on while tote floor is still visible in a cell.
- **SOP rule broken:** Steps 1 to 4, the order never changes.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** a gripper does the work of the gripper on the other side: it sets the divider, lines a cell, takes a glass, or picks up a void pad on the other side of the surface, or fills the other void itself instead of handing the pad over; or the left gripper picks up the sticker.
- **SOP rule broken:** Steps 1 to 4, each thing is taken by the gripper on its side, the pad gripper hands the other pad over, and the right gripper picks up the sticker. Adjusting or pushing something already in the tote is not a role: either gripper or both may do it.
- **Coaching note:** know each gripper's work before you start. Nothing crosses the middle.

Violation: Cells filled out of order

- **Visible cue:** a cell is lined or filled while a row behind it still has an empty cell, or the cell on the glass side of a row is filled before the cell away from it.
- **SOP rule broken:** Steps 2.1 and 2.2, the next cell is always the backmost empty one, the cell away from the glass side first.
- **Coaching note:** back row first, far cell first.

Violation: Divider skipped or badly set

- **Visible cue:** a sheet or a glass goes into a tote with no divider, or the divider sits crooked, rides up on a wall, or has a bent or folded wall when the first sheet goes in.
- **SOP rule broken:** Step 1, the divider is seated on the tote floor before anything else goes in.
- **Coaching note:** flat, square, every cell open, then start packing.

Violation: Cell not lined

- **Visible cue:** a glass goes into a cell with no sheet in it, or into a sheet that is crushed, folded in on a side, or caught on a wall short of the floor.
- **SOP rule broken:** Step 2.1, a sheet lines every cell, flat against all four walls, before its glass goes in.

- **Coaching note:** sheet first, then the glass. Push it down until it lines the cell.

Violation: Glass touched a glass or bare cardboard

- **Visible cue:** two glasses meet, a glass is lowered against the top of a divider wall, a glass rubs bare cardboard, or a glass rings against anything in the tote.
- **SOP rule broken:** Step 2.2, each glass goes straight down inside its sheet and lies flat on its side touching nothing but bubble wrap.
- **Coaching note:** line up over the cell before you go down. Glass never touches glass.

Violation: Broken glass packed

- **Visible cue:** a glass that is chipped, cracked, or broken is put into the tote, or a glass cracks or breaks while it is being packed and the episode goes on with it in the tote.
- **SOP rule broken:** Step 2.2, never pack a glass that is chipped, cracked, or broken, or one that cracks while it goes in.
- **Coaching note:** check each glass before it goes in. A damaged glass never goes in the tote.

Violation: Glass forced in

- **Visible cue:** a glass is dropped onto its cell, or pushed hard enough to crack, chip, or tip it.
- **SOP rule broken:** Steps 2.2 and 3, a glass is lowered in and pushed only gently to seat it.
- **Coaching note:** carry it any way you like, but bring it down gently.

Violation: Dropped or released over the wrong place

- **Visible cue:** a gripper opens above its spot and the glass, divider, sheet, void pad, or sticker falls the rest of the way; a glass is let go before it is resting on the tote floor; or a gripper opens over the surface, the pick row, or open space and the thing lands anywhere but the spot it was going to.
- **SOP rule broken:** Steps 1 to 4, lower in until it is resting over the release point, then let go, and open nowhere else.
- **Coaching note:** know where it lands, and take it all the way down before you open the gripper.

Violation: Void pad hand-over broken

- **Visible cue:** the other gripper reaches to the pad spot for its pad itself. The pad gripper opens before the other gripper has closed, or the pad passes with no half second where both hold it. The hand-over happens away from the hand-over place. Or the pad is dropped between the grippers.
- **SOP rule broken:** Step 3.2, the pad gripper carries the other void pad to the hand-over place, both hold it for half a second, then the pad gripper opens.

- **Coaching note:** carry it to the place, wait for the close, hold together, then let go.

Violation: Glass not lying flat

- **Visible cue:** a glass is set in its cell standing upright, propped up on its rim or its base, or still rolling or rocking when the gripper lifts clear, and the episode goes on.
- **SOP rule broken:** Step 2.2, every glass lies flat on its side and settled in its cell before the next thing goes in.
- **Coaching note:** watch it settle before you move on.

Violation: Voids left unfilled

- **Visible cue:** the sticker goes on with tote floor visible down either side of the divider, a pad sitting short of the void's full length, or a void pad still at the pad spot.
- **SOP rule broken:** Step 3, both voids are filled before the sticker goes on.
- **Coaching note:** check both sides of the divider. If floor shows, feed the pad down.

Violation: Sticker mishandled

- **Visible cue:** the sticker lands printed side down, inside the tote, on a glass, on a void pad, hanging off the ledge, or folded or stuck to itself when the episode ends; it ends on the label ledge upside down, turned on its side, or at any angle other than the right way up, so its words do not read left to right from the front of the surface; or the episode ends with the sticker still at the sticker spot.
- **SOP rule broken:** Step 4, the sticker ends stuck flat on the label ledge the right way up.
- **Coaching note:** peel by the tab, keep it the way it lay at the front center, lay it flat with both grippers, press both halves, and adjust it if it sits crooked. If it lands wrong, peel it up and lay it again.

Violation: More than one thing handled at once

- **Visible cue:** a gripper holds more than one thing at the same time, such as glasses, a glass and a sheet, or the void pads; or a gripper starts on the void pads while a glass is still being carried.
- **SOP rule broken:** Steps 1 to 4, each gripper holds only the thing it is working on, and each substep finishes before the next one starts.
- **Coaching note:** finish what the gripper is holding, then start the next.

Violation: Step skipped

- **Visible cue:** a step or substep is left out: the divider never goes in, a glass goes in with no sheet run for its cell, a void is never padded, the pad is never handed over, the sticker never goes on, or the operator moves on to a later step with an earlier one not done.
- **SOP rule broken:** Steps 1 to 5, every step and substep is run, in order.

- **Coaching note:** run every step, every time. If one was missed, go back and do it before moving on.

Violation: Tote or pick row moved

- **Visible cue:** the tote or a glass still in the pick row is pushed, dragged, lifted, or turned during the episode; the tote is pulled toward the front edge to bring it within reach; or a gripper leans on the tote rim hard enough to shift it.
- **SOP rule broken:** Steps 1 to 4, the tote and the pick row stay where they are put.
- **Coaching note:** reach out to it. Never bring it to you.

Violation: Released before settled

- **Visible cue:** a glass, the divider, a sheet, a void pad, or the sticker is let go before it rests in its spot and falls (dropped) onto the surface.
- **SOP rule broken:** Steps 1 to 4, let go only after a settled placement.
- **Coaching note:** check the landing spot before you let go.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a glass still in the pick row or on the surface, a glass in an unlined cell, a void unfilled, the sticker off the ledge, an arm away from home, or an arm still over the tote.
- **SOP rule broken:** Step 5, check the packed tote and the empty pick row; return both arms home clear of the scene; then stop recording.
- **Coaching note:** check first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Glass already damaged** when the step begins: chipped, cracked, starred, or unable to stand on its own base. Replace it before the next episode.
- **Two glasses cannot be told apart by weight** from their shape, so the next glass in cannot be chosen. Discard the episode and restock the pick row with glasses whose weights are told apart at a glance.
- **Wrap or paper missing or short:** the sheet row runs out of bubble wrap sheets before the pick row is empty, or a void pad is missing from the pad spot. Discard the episode and restock the sheets and pads.

- **Sheet will not stand** on its own on a flat surface even after it is refolded. Replace it before the next episode.
- **Divider does not fit:** it will not drop into the tote and sit flat, a wall is bent, or it has fewer cells than there are glasses in the pick row. Discard the episode and replace the divider or restock the pick row.
- **Tote fault:** the tote is cracked, will not sit square, or its label ledge will not take the sticker flat. Replace it before the next episode.
- **Fragile sticker missing, blank, unreadable, or no longer sticky** when Step 4 begins.
- **Tote or glass slides** under a correct lower in because the surface is slick, wet, or crumbed. Wipe and dry the surface before the next episode.

Annotation subtasks (from SOP)

1. Set the cell divider in the tote
2. Line one cell with a bubble wrap sheet
3. Move one glass from the pick row into its cell
4. Fill one void with a void pad
5. Pass the other void pad across to the other gripper
6. Stick the fragile sticker on the tote ledge
7. End the episode